

Renal System

The Structure of the renal system:

The renal system consists of:

- 1) 2 kidneys.
- 2) 2 ureters.
- 3) Urinary bladder.

Functions of Kidneys:

1. Formation of urine (**homeostasis**)

(Maintain fluid-electrolyte and acid-base balance-
Eliminate wastes and excess ions)

2. Endocrine functions:

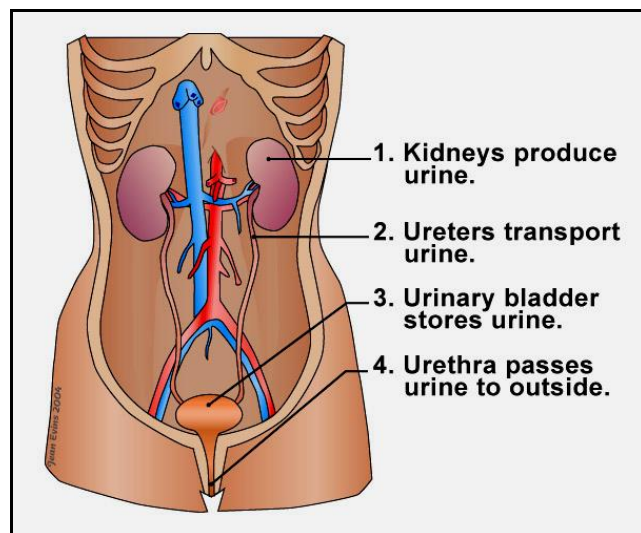
A- Synthesize, store and release of **Renin**

by **Juxta glomerular (JG) cells** (which governs
blood pressure and kidney function)

B- Erythropoietin hormone that stimulates red blood cell production.

3. Activation of **Vitamin D**. active form of vitamin D, 1,25-dihydroxycholecalciferol induces the synthesis of an intestinal Ca^{2+} -binding protein, calbindin D-28K.

■ Vitamin D deficiency or chronic renal failure results in inadequate intestinal Ca^{2+} absorption, causing rickets in children and osteomalacia in adults.



The kidney consists of:

- 1) Cortex.
- 2) Medulla.

It contains functional unit called the nephron.

The Nephron:

- The Nephron is the functional unit of the kidney.
- Each kidney consists of one million of nephrons.

The Nephron consists of:

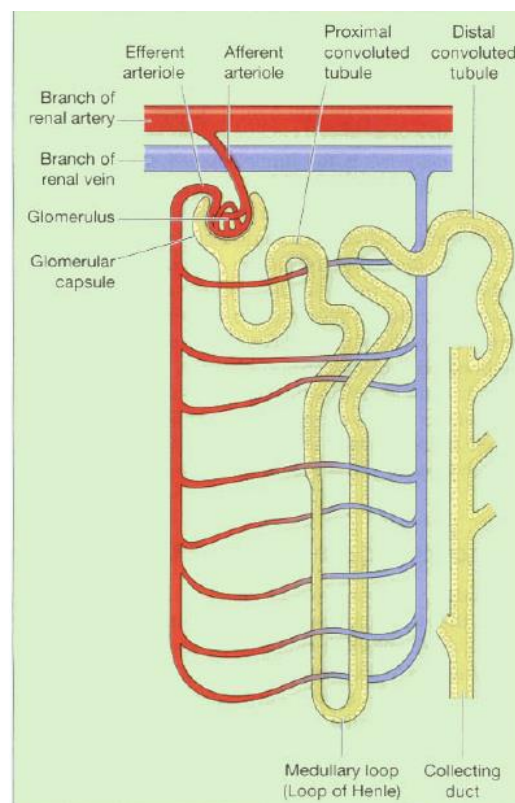
I- Glomerulus.

II- Renal Tubules: They are proximal convoluted tubules, loop of henle and distal convoluted tubules.

Formation of urine:

- The urine is formed by combination of the following processes:

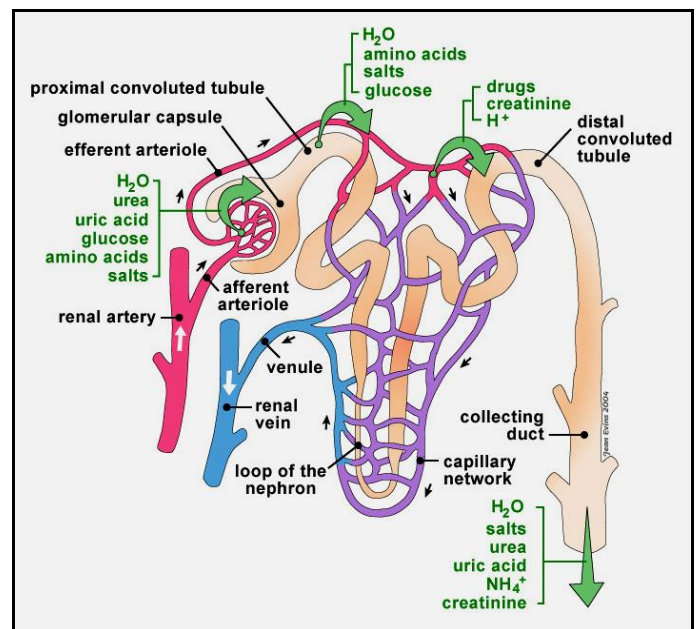
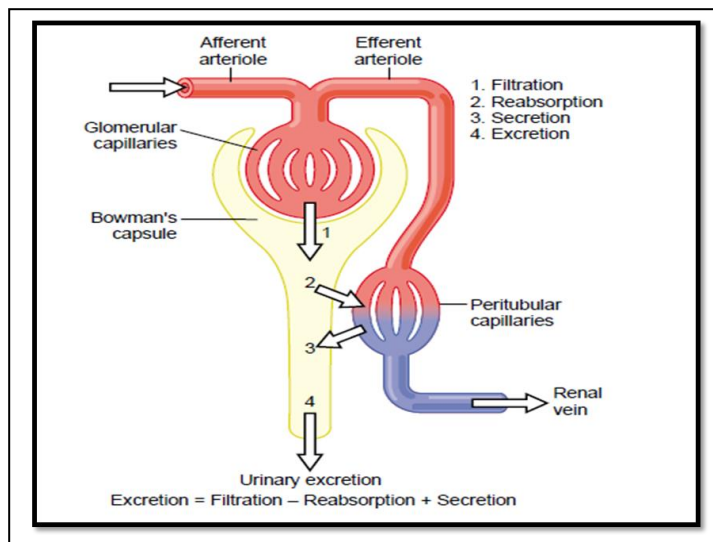
- 1- Glomerular filtration.
- 2- Tubular reabsorption.
- 3- Tubular secretion.



Urine Formation by Nephron

- 1- Glomerular filtration: blood pressure forces glucose, amino acids, uric acid, urea, H₂O from the glomerulus to the capsule
2. Tubular Reabsorption: return of filtrates from blood at the proximal tubule through *diffusion and active transport* (glucose, amino acids, salts and H₂O)
3. Tubular Secretion: movement of drugs and toxins from blood into the distal convoluted tubule
4. Reabsorption of water: return of H₂O via osmosis along the loop of Henle and collecting duct.

Urinary excretion = Glomerular Filtration – Tubular Reabsorption + Tubular Secretion.
The following figure showing these processes



- Urine moves from the collecting ducts through the kidney *pelvis* to the *ureter*
- Urine moves from the kidneys, through the *ureters* to the bladder and finally through the *urethra*

Renal blood flow

- Renal blood flow = 1200 ml blood/ minute.
Renal plasma flow = 650 ml plasma/ min.
So, Renal fraction of cardiac output = $\frac{1200}{5000} = 25\%$ of C OP.

Glomerular Filtration:

Glomerular filtrate (GF:) is the fluid filtered from plasma through the glomerular membrane and Bowman's capsule into the renal tubule.

Glomerular filtration rate (GFR):

- It is the amount of glomerular filtrate formed in all nephrons of both kidneys /minute.
- In normal adult human, GFR averages 125 ml/min = 180 liters/day.

Dynamics of Glomerular Filtration (GF)

Forces favoring glomerular filtration:

1. *Glomerular hydrostatic pressure*: Normally 60 mm Hg in humans.
2. *Colloid osmotic pressure of proteins in Bowman's capsule*: zero because normally no proteins can pass into the GF.

Forces opposing filtration:

1. *Colloid osmotic pressure of plasma proteins*, with an average value 32 mm Hg.
2. *Hydrostatic pressure in Bowman's capsule*: It is about 18 mmHg.

Net filtration pressure = $(60+0) - (32+18) = +10 \text{ mmHg}$.

Factors affecting GFR: these factors include the following:

A) Glomerular hydrostatic pressure:

Increased glomerular hydrostatic pressure $\longrightarrow$ increased filtration pressure $\longrightarrow$ increased GFR, and vice versa.

It is affected by:

1- Afferent arteriolar resistance:

- Vasoconstriction of afferent arterioles $\rightarrow$ decreased glomerular hydrostatic pressure $\rightarrow$ decreased GFR, and vice versa

2- Efferent arteriolar resistance:

- Vasoconstriction of efferent arterioles $\rightarrow$ increased glomerular hydrostatic pressure $\rightarrow$ increased GFR.

B) Plasma proteins colloid osmotic pressure:

Increased arterial plasma colloid osmotic pressure $\longrightarrow$ decreased GFR and vice versa.

C) Hydrostatic pressure in Bowman's capsule:

Increased hydrostatic pressure in Bowman's capsule e.g. (due to ureteric stones) $\longrightarrow$ decreased GFR.

Functions of Renal Tubules

Factors affecting tubular Reabsorption: these factors include the following:

I) Physical Forces across peri-tubular Capillaries:

A) Forces helping absorption:

- 1- Colloid osmotic pressure of peri-tubular capillary plasma proteins... 32 mmHg.

2-Hydrostatic pressure in the renal interstitium 6 mmHg.

Total reabsorption force = 38 mmHg

B) Forces opposing absorption :

1- Hydrostatic pressure in the peritubular capillaries.... 13 mmHg.

2-Colloid osmotic pressure of proteins in renal interstitium....15 mmHg.

Total = 28 mmHg.

Net reabsorptive force = 38 - 28 = 10 mmHg.

II) Hormonal Control:

1) Aldosterone:

-Is secreted by *adrenal cortex*. It acts mainly on distal and cortical collecting tubules and *increased reabsorption of sodium and water*.

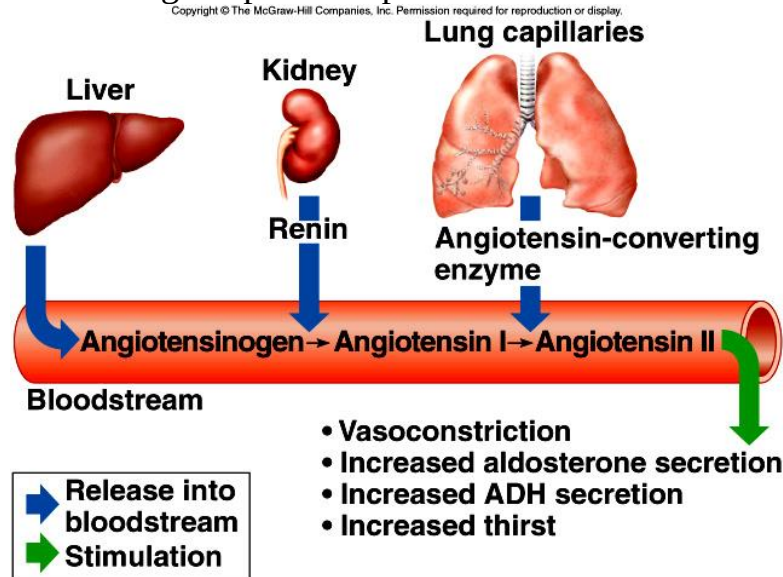
2) Angiotensin II:

- When ECF volume and ABP decreased (as in hemorrhage) → increased renin secretion and angiotensin II formation →

1- *Increased aldosterone secretion*.

2- *Constriction of efferent arterioles*.

The following figure showing the previous process



3)Antidiuretic hormone (ADH):

It increases reabsorption of water by the distal tubule and collecting duct.

4) Atrial natruretic peptide (ANP):

-Increased blood volume ----- distention of cardiac atria ----- increased ANP secretion ----- decreased sodium and water reabsorption by renal tubule ----- decrease of blood volume toward normal.

Functions of proximal convoluted tubules:

1. Absorption of all filtered glucose, aminoacids, proteins and vitamins.
2. Absorption of 65% of filtered load of Na, Ca, Cl and bicarbonate.
3. Absorption of 65% of filtered water.

4. Secretion of:
 - bile salts, oxalate, urates.
 - waste products: urea and creatinin.

Functions of loop of Henle:

1. 20 % of the filtered water is reabsorbed in the loop of Henle.
2. Absorption of 25% of filtered load of Na, Cl and K.
3. Secretion of H ions.

MICTURITION

Definition:

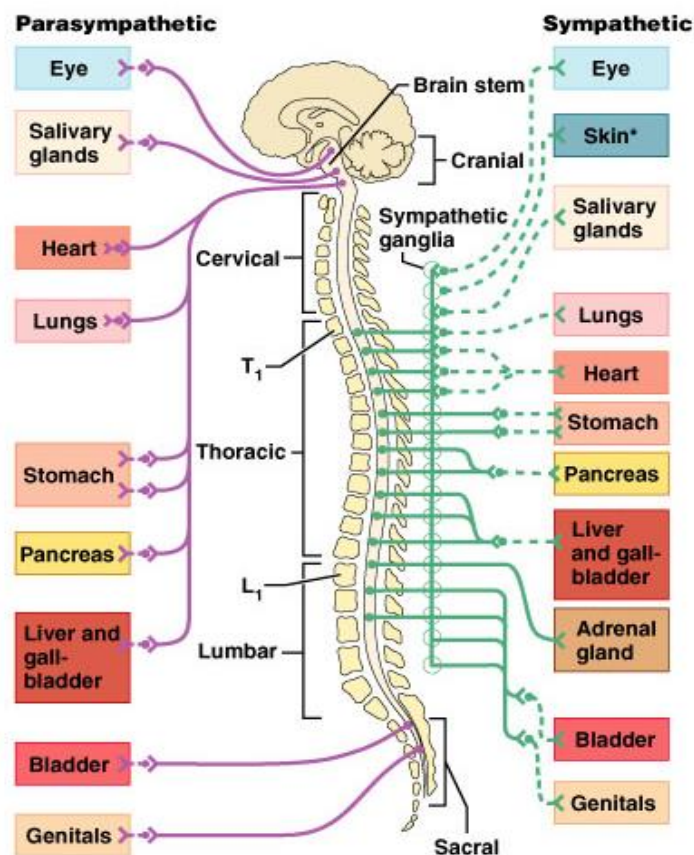
is the process by which the urinary bladder empties when it becomes filled.

Innervation of the urinary bladder:

Autonomic fibers:

Parasympathetic nerve fibers:

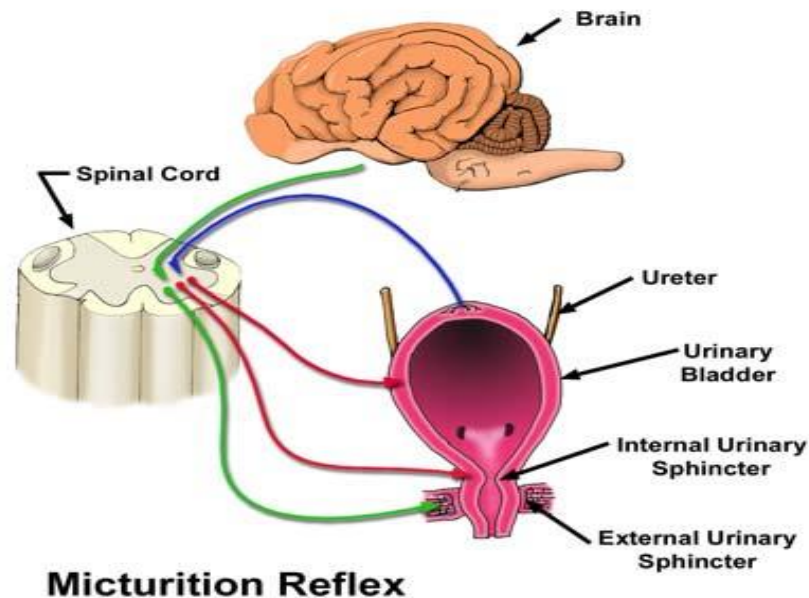
- 2 , 3 and 4 sacral segments of the spinal cord → run in the pelvic nerve .
- They relay in terminal ganglia in the wall of the urinary bladder. The following diagram showing this innervation.



Micturition reflex:

It is a spinal reflex:

Distension of the bladder with urine → afferent impulses to S 2 & 3 & 4 segments of spinal cord → Efferent impulses form the center along the pelvic nerve → contraction of the wall of the bladder and relaxation of the *internal urethral sphincter*.



Brain control of micturition:

1- In-between urination

- a) Keeping the micturition reflex partially inhibited.
- b) Continuous contraction of the *external urethral sphincter*.

2- At time of urination:

- a) Facilitates the sacral micturition center to initiate micturition reflex.
- b) Inhibits the *external urethral sphincter*.